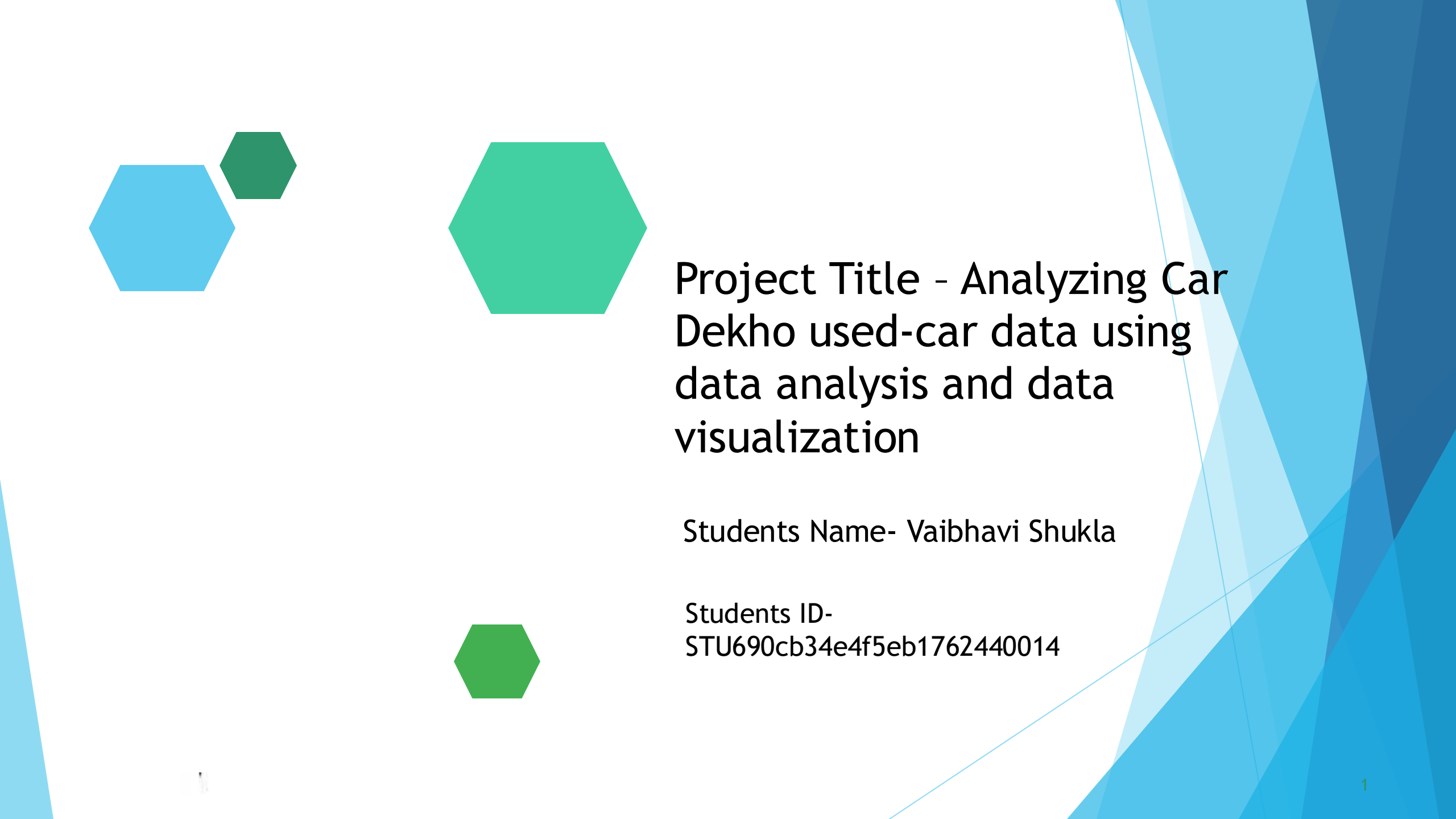


The slide features a white background with several decorative elements. On the left, there are three hexagons: a light blue one, a dark green one, and a medium green one. In the center, there is a large medium green hexagon and a smaller green hexagon at the bottom. On the right, there is a large, abstract, multi-colored blue shape that resembles a stylized 'C' or a series of overlapping triangles. The text is positioned to the right of the central hexagons.

Project Title - Analyzing Car Dekho used-car data using data analysis and data visualization

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Students ID-
STU690cb34e4f5eb1762440014

PROBLEM STATEMENT

- 1. The used-car market contains vehicles with different prices, fuel types, transmission types, and usage levels.
- 2. It is difficult to understand which factors have the greatest impact on a car's selling price.
- 3. Buyers and sellers need data-driven insights to evaluate a vehicle's fair market value.
- 4. This project analyzes historical Car Dekho data to identify important market trends and pricing patterns.
- 5. The goal is to provide meaningful insights that can support better buying, selling, and pricing decisions.



Project Description

This project focuses on analyzing Car Dekho used-car data using data analysis and visualization techniques. The dataset contains information such as car name, manufacturing year, selling price, present price, kilometers driven, fuel type, seller type, transmission, and number of previous owners. The analysis explores relationships between these variables and identifies the major factors influencing used-car prices. The project can help stakeholders understand market trends and make informed decisions.

End Users

- Car Buyers - to compare cars and identify reasonable prices.
- Car Sellers - to estimate competitive selling prices.
- Car Dealers - to understand demand and pricing trends.
- Used-Car Businesses - to support inventory and pricing decisions.
- Data Analysts/Business Teams - to derive insights from automobile market data.

WHO ARE THE END USERS?

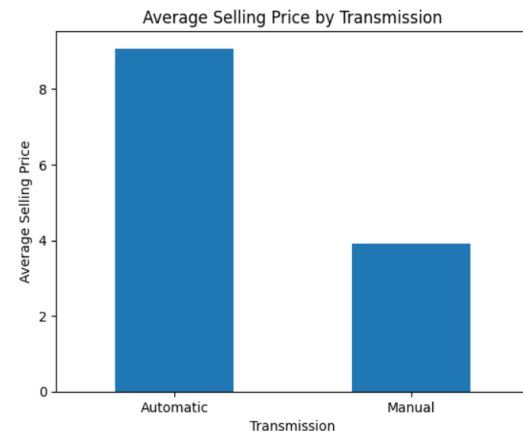
- The primary end users of the Car Market Trends Analysis system are car buyers, used-car dealers, automobile companies, online automobile platforms, data analysts, and researchers. The analysis helps these users understand market trends, compare prices, identify popular car categories, and make data-driven decisions regarding buying, selling, and pricing of used cars.

Technology Used

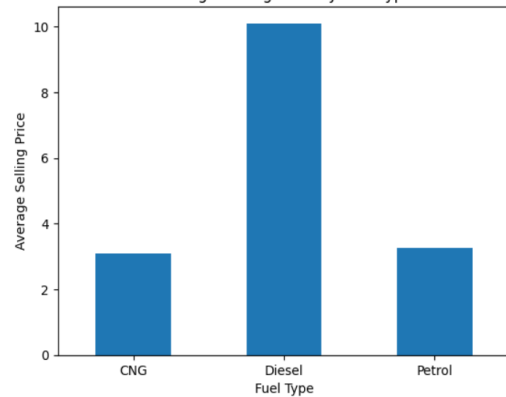
- Technology Used : Python , Pandas, Matplotlib, Seaborn and Num Py



plt.show()



Average Selling Price by Fuel Type



```
[2] ✓ 0s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
[16] ✓ 0s
df = pd.read_csv('/content/1776311302-P3-Car Market Trends Analysis with Car Dekho Data (1).csv')
```

```
[17] ✓ 0s
df.head()
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[18] ✓ 0s
df.shape
```

```
df.describe(include='all').T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Car_Name	301	98	city	26	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Year	301.0	NaN	NaN	NaN	2013.627907	2.891554	2003.0	2012.0	2014.0	2016.0	2018.0
Selling_Price	301.0	NaN	NaN	NaN	4.661296	5.082812	0.1	0.9	3.6	6.0	35.0
Present_Price	301.0	NaN	NaN	NaN	7.628472	8.644115	0.32	1.2	6.4	9.9	92.6
Kms_Driven	301.0	NaN	NaN	NaN	36947.20598	38886.883882	500.0	15000.0	32000.0	48767.0	500000.0
Fuel_Type	301	3	Petrol	239	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Seller_Type	301	2	Dealer	195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Transmission	301	2	Manual	261	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Owner	301.0	NaN	NaN	NaN	0.043189	0.247915	0.0	0.0	0.0	0.0	3.0

RESULTS

Demo Link-<https://github.com/vaibhavishukla45-code/VOIS-.git>

Thank you